

Information Technology University (ITU).
CP_Submission-1: Code Project Proposal

VisiSlide: A Multimodal Gesture and Attention-Driven Presentation Interface

Subject: Human computer interaction

Assignment 1

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1. Research Project Title

VisiSlide: A Multimodal Gesture and Attention Driven Presentation Interface

VisiSlide is an intent-aware, hands free presentation tool designed to make digital speaking feel more natural and fluid. Instead fumbling with a clicker or a mouse, you can change slides using simple hand gestures but only when you're actually looking at the camera(active attention). By, combining the Media-pipe based sensing along with a deterministic based verification layer, the system ensures that your slides move only when you want them to(looking at the camera).

2. Problem Statement

Traditional presentations often feel disconnected. Whether it's a lecturer stuck behind a podium or a speaker constantly looking down at their laptop, the technology often gets in the way. We are specifically solving three core HCI issues:

Key issues include:

- **The Midas Touch Problem:** Most gesture only systems based on 100% ML are too sensitive accidentally switching slides and cannot differentiate between natural hand-talking or deliberate command.
 - **Lost Connection:** Speakers often lose eye contact with their audience because they have to keep checking their screen.
 - **Accessibility:** Clickers and mice aren't an option for everyone especially presenters with motor disabilities who may find it difficult to grip hardware for long periods.
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3. Target Users

We designed VisiSlide for anyone who wants to command a room without touching a button:

- **Teachers & Researchers:** To allow them to freely circulate the classroom and maintain eye contact with students.
 - **Corporate Pros:** For those crucial pitches where you want the technology to be seamless and undetectable.
 - **Virtual Speakers:** In order to facilitate slide navigation for online presenters who are unable to turn away from their webcam.
 - **Inclusive Design:** To offer a dependable device to people who are unable to utilize conventional physical remotes.
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4. Interaction Challenges

Building this isn't just about recognizing a hand it is about recognizing intent. Our biggest hurdles include:

- **Multimodal Logic:** Creating a Gaze Gate where a gesture is only accepted if the user's gaze is verified otherwise they will be ignored mathematically.
 - **Real-World Environments:** Making sure the webcam can still see you with high-fidelity landmark detection in poorly lit auditoriums or bright classrooms (low-fidelity environment).
 - **Zero Calibration:** Unlike most eye trackers that require a long setup, we want a plug-and-play experience using MediaPipe based landmarks.
 - **Instant Feedback:** Ensuring there is no lag by eliminating black-box feeling of AI and providing instant visual to the user, so the speaker feels in total control.
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5. Motivation & significance

The primary goal of the VisiSlide is to remove the hurdles between the presenter and their digital media. In the current ages or settings, the technology often works as the physical chain – That forces the lecturers(presenters) to stand behind the dais podium for operating the mouse. The next frontier in HCI according to the study **Korkmaz (2026)** is not just improving sensing but also the development of the “enabling environments” that facilitate natural interaction and accessibility.

While Mohamed (2025) proved that the eye gesture can reach near-perfect accuracy, our project uses this to lessen the chronic mental load of the presenters. By removing the need of holding hardware in the hands, we provide an amazing and user-friendly tool to the speakers with neuromuscular deficits, and simultaneously it allows the robust personalities to regain their stage presence. The actual goal is to make computer hidden partner that understands the user cues and works accordingly.

6. Existing System Limitations

In the Current era the gesture-based systems are struggling with a gap known as “Intent-Gap”. Large number of systems described by **Li-Hsieh (2025)** are actually the classification models and working on the 100% ML

The “Midas-Touch” conflict: In professional speaking, people do talk with their hands to align the audience with them. But the 100% ML based systems translate that body language into the command and began to perform operation like zooming in the slide or moving on to the next slide (embarrassing interruptions).

Stochastic Behavior: 100% ML systems are actually the “Black-Boxes” the user usually feels the lack of sense of control. If any of the gesture fails the user will lose his confidence because he will not actually know the reason of the embarrassing behavior of the system.

Hardware Dependency: According to **Tejaswini & Rao's (2025)** examination of the current solutions, they call for specific kinds of clicking devices and infrared sensors.

The solutions that are based upon webcam fail in the low-fidelity environment like the dim lights of the auditorium.

Social Disconnect: When the presenter has to look at his laptop screen for controlling the movement of slides or for some other purpose it results in the loss of eye contact with the audience and sets a very bad impact.

7. Proposed solution:

VisiSlide instigates a **Hybrid Intent-Gating Framework** that is not actually based on the "Pure ML" but on "**Human-Centric Heuristics**." The introduced system is not just limited to detecting the gesture; it adds on the verification of intent behind the gesture.

Our introduced system merges the amazing sensing capabilities of **Media Pipe** with a **Gaze-Driven Logic Gate**. The interaction logic is distributed in three layers discussed below:

1. **Sensing Layer (ML):** The technique gathers 3D landmarks for the hand skeletons and face mesh using a webcam.
2. **The Gaze Gate (HCI Heuristic):** This is actually the core of Machine Learning. Instead of an ML, the system uses a **deterministic mathematical check**. It calculates the Euclidean distance and vector angle between the user's pupils and the camera lens. A gesture will only be "unlocked" if the user's gaze is verified to be on the camera.
3. **Temporal Verification:** In line with **Li & Hsieh's (2025)** research, we prioritize dynamic swipes over static stances. Only intentional, fast "flicks" cause a slide shift, according to our analysis of the hand's trajectory over a 30-frame span.

So, the VisiSlide ensures that you can talk freely with your audience without taking care about controlling your Slide. It will save you from the embarrassing moment because the gestures will unlock when you have intentions to unlock them.

References & Links

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 2. **Rajanala, V. P., et al. (2026).** *Hand Gesture Controlled Presentation System.*
Focus: Uses OpenCV and MediaPipe for real-time slide navigation.
 - [Read Paper at IJRASET](#)
 3. **MDPI Research (2026).** *Mapping Eye-Tracking Research in HCI.*
Focus: Explores eye-tracking as a tool for reducing cognitive load.
 - [Read Paper at MDPI](#)
 4. **Malleswara Rao, K. (2025).** *Dynamic Hand Gesture Recognition Using Transformer Networks.*
Focus: Using temporal sequence (swipes) rather than static poses to avoid "Midas Touch."
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